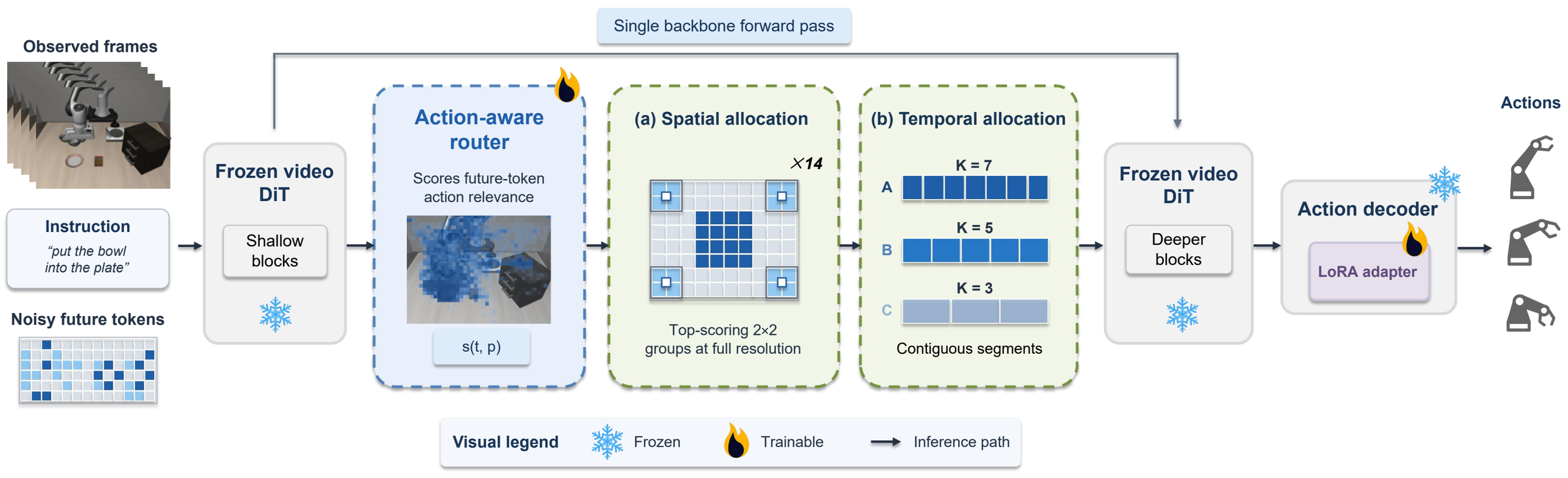

Single backbone forward pass
Observed frames
Instruction
"put the bowl into the plate"
Noisy future tokens
Frozen video DiT
Shallow blocks
Action-aware router
Scores future-token action relevance
s(t, p)
(a) Spatial allocation
×14
Top-scoring 2×2 groups at full resolution
(b) Temporal allocation
A
K = 7
B
K = 5
C
K = 3
Contiguous segments
Frozen video DiT
Deeper blocks
Action decoder
LoRA adapter
Actions
Visual legend
Frozen
Trainable
Inference path